

# Supervised Bayesian Matrix Factorization

V2 Companion Document: Derivations & Code Mapping

Reference for `Supervised_Bayesian_MF_V2.R`

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March 2026

## Abstract

This companion document provides a complete, self-contained derivation of the Supervised Bayesian Matrix Factorization model and maps every mathematical quantity to its corresponding variable in the V2 R implementation. It is designed to be read alongside `code/Supervised_Bayesian_MF_V2.R` and serves as the authoritative reference for both the theory and its computational realisation. All corrections from the derivation review (R1–R8, documented in `MF_Derivations_UpdateAlgo_REVISED.tex`) are incorporated throughout.

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# 1 Model Specification & Notation

## 1.1 Index Conventions

| Symbol            | Dimension    | Meaning                                | R Variable             |
|-------------------|--------------|----------------------------------------|------------------------|
| $n$               | scalar       | number of samples (patients)           | <b>n</b>               |
| $p$               | scalar       | number of features (genes / CpG sites) | <b>p</b>               |
| $K$               | scalar       | number of latent factors               | <b>K</b>               |
| $i$               | $1 \dots n$  | sample index                           | row index              |
| $j$               | $1 \dots p$  | feature index                          | column index           |
| $k, k'$           | $1 \dots K$  | factor index                           | loop variable <b>k</b> |
| $Y$               | $n \times p$ | observed genomics data matrix          | <b>Y</b>               |
| $L$               | $n \times K$ | loading matrix                         | <b>EL</b>              |
| $F$               | $p \times K$ | factor matrix                          | <b>EF</b>              |
| $\beta$           | $K \times 1$ | survival coefficient vector            | <b>EBeta</b>           |
| $\tau_j$          | $p$ -vector  | feature-specific noise precision       | <b>Tau</b>             |
| $(t_i, \delta_i)$ | $n$ -vectors | survival time & event indicator        | <b>time, status</b>    |

## 1.2 Posterior Moment Notation

Throughout this document,  $q(\cdot)$  denotes the variational posterior and  $g(\cdot)$  denotes the (learned) prior. The critical distinction is between the *posterior mean* and the *posterior second moment*:

$$\bar{l}_{ik} := \mathbb{E}_q[l_{ik}] \quad (\text{posterior mean}) \quad (1)$$

$$\overline{l_{ik}^2} := \mathbb{E}_q[l_{ik}^2] = \text{Var}_q(l_{ik}) + \bar{l}_{ik}^2 \quad (\text{posterior 2nd moment}) \quad (2)$$

and analogously for  $\bar{f}_{jk}$ ,  $\overline{f_{jk}^2}$ ,  $\bar{\beta}_k$ ,  $\overline{\beta_k^2}$ . Whenever there is posterior uncertainty, the second moment strictly exceeds the squared mean:  $\overline{l_{ik}^2} \geq \bar{l}_{ik}^2$ .

### Code Mapping (V2.R)

| Math                   | R Variable       | Computation in V2.R                       |
|------------------------|------------------|-------------------------------------------|
| $\bar{l}_{ik}$         | <b>EL[i,k]</b>   | <b>res_L\$posterior\$mean</b>             |
| $\overline{l_{ik}^2}$  | <b>EL2[i,k]</b>  | <b>res_L\$posterior\$sd^2 + mean^2</b>    |
| $\bar{f}_{jk}$         | <b>EF[j,k]</b>   | <b>res_F\$posterior\$mean</b>             |
| $\overline{f_{jk}^2}$  | <b>EF2[j,k]</b>  | <b>res_F\$posterior\$sd^2 + mean^2</b>    |
| $\bar{\beta}_k$        | <b>EBeta[k]</b>  | <b>res_Beta\$posterior\$mean</b>          |
| $\overline{\beta_k^2}$ | <b>EBeta2[k]</b> | <b>res_Beta\$posterior\$sd^2 + mean^2</b> |

## 1.3 Genomics Model

The observed data matrix  $Y$  is generated by a low-rank signal plus element-wise Gaussian noise:

$$Y_{n \times p} = L_{n \times K} F_{K \times p}^\top + E_{n \times p}, \quad E_{ij} \sim \mathcal{N}(0, \tau_j^{-1}) \quad (3)$$

Element-wise:  $Y_{ij} = \sum_{k=1}^K l_{ik} f_{jk} + \varepsilon_{ij}$ . The noise precision  $\tau_j$  is feature-specific (column-specific), allowing heterogeneous noise levels across genomic features.

## 1.4 Survival Model (Cox Proportional Hazards)

Patient survival is modelled via a Cox PH model that links the latent loadings  $\mathbf{l}_i = (l_{i1}, \dots, l_{iK})^\top$  to the hazard:

$$h(t_i) = h_0(t_i) \exp(\eta_i), \quad \eta_i = \mathbf{l}_i^\top \boldsymbol{\beta} = \sum_{k=1}^K l_{ik} \beta_k \quad (4)$$

where  $h_0$  is an unspecified baseline hazard and  $\boldsymbol{\beta}$  quantifies the prognostic contribution of each latent factor.

## 1.5 Hierarchical Priors

All latent variables receive empirical Bayes priors from the EBNM framework:

$$l_{ik} \stackrel{\text{iid}}{\sim} g_l \in \mathcal{G}_l, \quad f_{jk} \stackrel{\text{iid}}{\sim} g_f \in \mathcal{G}_f, \quad \beta_k \stackrel{\text{iid}}{\sim} g_\beta \in \mathcal{G}_\beta$$

In V2.R, all three use `prior_family = "point_normal"`, i.e., a mixture of a point mass at zero and a normal distribution with data-adaptive variance. This promotes sparsity in all three parameter sets.

# 2 Evidence Lower Bound (ELBO)

## 2.1 Variational Family

We adopt a mean-field variational family:

$$q(L, F, \boldsymbol{\beta}) = \prod_{k=1}^K q_{l_k}(\mathbf{l}_{\cdot k}) \prod_{k=1}^K q_{f_k}(\mathbf{f}_{\cdot k}) \prod_{k=1}^K q_{\beta_k}(\beta_k) \quad (5)$$

## 2.2 ELBO Expression

The ELBO to be *maximised* over posteriors  $(q_l, q_f, q_\beta)$ , priors  $(g_l, g_f, g_\beta)$ , and precision  $\tau$  is:

$$\begin{aligned} \mathcal{F} = & \underbrace{\mathbb{E}_q[\log P(Y \mid L, F, \tau)]}_{\text{genomics log-lik}} + \underbrace{\mathbb{E}_q[\log P(t, \delta \mid L, \boldsymbol{\beta})]}_{\text{survival log-lik}} \\ & + \sum_k \mathbb{E}_{q_{l_k}} \left[ \log \frac{g_{l_k}}{q_{l_k}} \right] + \sum_k \mathbb{E}_{q_{f_k}} \left[ \log \frac{g_{f_k}}{q_{f_k}} \right] + \sum_k \mathbb{E}_{q_{\beta_k}} \left[ \log \frac{g_{\beta_k}}{q_{\beta_k}} \right] \end{aligned} \quad (6)$$

The key structural property is conditional independence:

$$\log P(Y, t, \delta \mid L, F, \boldsymbol{\beta}, \tau) = \log P(Y \mid L, F, \tau) + \log P(t, \delta \mid L, \boldsymbol{\beta}) \quad (7)$$

This means the genomics and survival likelihoods contribute additively, and each coordinate update can separately aggregate terms from both sources.

#### Code Mapping (V2.R)

V2.R tracks the *genomics* portion of the ELBO as a monitoring diagnostic (the survival portion is approximated via Taylor expansion and does not have a simple closed-form):

```
# Line 379: ELBO proxy = E_q[log P(Y | L, F, tau)]  
history$elbo_proxy[iter] <- sum(n/2 * log(Tau) - Tau/2 * colSums(R2_bar))
```

This should be non-decreasing across iterations (monotonicity of CAVI for the *genomics* sub-problem).

### 3 Cox Partial Likelihood — Taylor Expansion

The Cox partial log-likelihood is non-conjugate in both  $L$  and  $\beta$ . To handle it within the EBNM framework, we use a second-order Taylor expansion to obtain a locally-Gaussian weighted least squares surrogate.

#### 3.1 Expansion Point

At each outer CAVI iteration, we fix the expansion point at the current posterior means:

$$\hat{\eta}_i = \sum_{k=1}^K \bar{l}_{ik} \bar{\beta}_k \quad (8)$$

#### Code Mapping (V2.R)

```
# Line 254: eta = EL %*% EBeta (n-vector)  
eta <- as.vector(EL %*% EBeta)
```

#### 3.2 Score and Negative Hessian

The Cox score (gradient) and negative diagonal Hessian at  $\hat{\eta}$  are:

$$u_i = \delta_i - \exp(\hat{\eta}_i) \sum_{j: t_j \leq t_i} \frac{\delta_j}{\sum_{m: t_m \geq t_j} \exp(\hat{\eta}_m)} \quad (\text{score}) \quad (9)$$

$$W_{ii} = \exp(\hat{\eta}_i) \sum_{j: t_j \leq t_i} \frac{\delta_j}{\sum_{m: t_m \geq t_j} \exp(\hat{\eta}_m)} \quad (\text{neg-diag Hessian, positive}) \quad (10)$$

Note that  $W_{ii} > 0$  for all observed samples, which ensures the quadratic surrogate is concave.

#### Code Mapping (V2.R)

```
# Lines 70-93: calc_cox_taylor(eta, time, status)  
# Returns: list(u = n-vector score, w = n-vector neg-diagonal Hessian)  
theta <- exp(eta_s) # exp(eta_i)  
risk_sum <- rev(cumsum(rev(theta))) # sum_{m: t_m >= t_i} exp(eta_m)  
h <- status_s / risk_sum # hazard increment
```

```

H <- cumsum(h)           # cumulative hazard
u_s <- status_s - theta * H # score
w_s <- theta * H         # neg-diag Hessian
w_s[w_s < 1e-6] <- 1e-6  # floor to prevent 0-division

```

The sorting by time (`ord <- order(time)`) is required because the at-risk set  $\{m : t_m \geq t_j\}$  must be computed via reverse cumulative sums over time-ordered samples.

### 3.3 Working Response

The 2nd-order Taylor expansion yields:

$$\ell(\boldsymbol{\eta}) \approx \sum_i \left[ -\frac{1}{2} W_{ii} (\eta_i - z_i)^2 \right] + C \quad (11)$$

where the **working response** is:

$$z_i = \hat{\eta}_i + \frac{u_i}{W_{ii}} \quad (12)$$

This has the form of a weighted Gaussian log-likelihood: response  $z_i$ , precision  $W_{ii}$ , mean  $\eta_i = \sum_k l_{ik} \beta_k$ . The IRLS (Iteratively Reweighted Least Squares) interpretation is:  $z_i$  is the “adjusted dependent variable” and  $W_{ii}$  is the “working weight.”

#### Code Mapping (V2.R)

```

# Lines 255-257: working response and weights
taylor <- calc_cox_taylor(eta, time, status)
z      <- eta + taylor$u / taylor$w # n-vector: z_i
w      <- taylor$w                 # n-vector: W_{ii}

```

These are held **fixed** for the entire inner  $k$ -loop (standard IRLS-within-VI). If `refresh_taylor = TRUE`, they are recomputed at each  $k$  (lines 268–273), giving a “true CAVI” at the cost of extra computation.

## 4 Update for $q_{l_k}$ — Patient Loadings

### 4.1 Partial Residuals

To isolate factor  $k$  in the coordinate-ascent update, we define:

**Genomics partial residual:**

$$R_{ij}^{-k} = Y_{ij} - \sum_{k' \neq k} \bar{l}_{ik'} \bar{f}_{jk'} \quad (13)$$

Equivalently,  $R^{-k} = Y - \bar{L} \bar{F}^\top + \bar{l}_k \bar{f}_k^\top$  (subtract the full reconstruction, then add back factor  $k$ ).

Survival partial working response:

$$z_i^{-k} = z_i - \sum_{k' \neq k} \bar{l}_{ik'} \bar{\beta}_{k'} \quad (14)$$

so that  $z_i \approx z_i^{-k} + l_{ik} \beta_k$ .

**Critical observation:**  $z_i^{-k}$  depends *only* on  $k' \neq k$  quantities — it does not depend on  $l_{ik}$  or  $\beta_k$ . This means the **same**  $z_i^{-k}$  can be reused for both the  $L$ -update and the  $\beta$ -update of factor  $k$  (see Section 6).

#### Code Mapping (V2.R)

```
# Lines 290-295: partial residuals inside k-loop
Y_hat <- EL %*% t(EF)                # n x p
R_k   <- Y - Y_hat + outer(EL[,k], EF[,k]) # R^{-k}

eta_no_k <- as.vector(EL %*% EBeta) - EL[,k] * EBeta[k]
z_no_k   <- z - eta_no_k                # z^{-k}
```

[A1] **Gauss-Seidel:** Because **EL** and **EBeta** are updated in-place during the  $k$ -loop, the partial residuals for factor  $k$  reflect updates from factors  $1, \dots, k-1$  within the same iteration.

## 4.2 Genomics Likelihood Contribution

Expanding the genomics log-likelihood around factor  $k$ :

$$\mathbb{E}_{q_f}[\log P(Y \mid L, F, \tau)] \supseteq \sum_{ij} \left[ -\frac{1}{2} \tau_j \left( R_{ij}^{-k} - l_{ik} f_{jk} \right)^2 \right] \quad (15)$$

$$= \sum_j -\frac{1}{2} \tau_j \mathbb{E}_q[f_{jk}^2] l_{ik}^2 + \sum_j \tau_j R_{ij}^{-k} \mathbb{E}_q[f_{jk}] l_{ik} + \text{const} \quad (16)$$

Collecting the coefficients of  $l_{ik}^2$  and  $l_{ik}$ :

$$A_{ik}^{\text{gen}} = \sum_{j=1}^p \tau_j \overline{f_{jk}^2} \quad (17)$$

$$B_{ik}^{\text{gen}} = \sum_{j=1}^p \tau_j R_{ij}^{-k} \bar{f}_{jk} \quad (18)$$

## 4.3 Survival Likelihood Contribution

From Eq. (11), isolating the terms in  $l_{ik}$ :

$$-\frac{1}{2} W_{ii} \left( z_i^{-k} - l_{ik} \beta_k \right)^2 = -\frac{1}{2} W_{ii} \mathbb{E}_q[\beta_k^2] l_{ik}^2 + W_{ii} z_i^{-k} \mathbb{E}_q[\beta_k] l_{ik} + \text{const} \quad (19)$$

The coefficients are:

$$A_{ik}^{\text{surv}} = W_{ii} \overline{\beta_k^2} \quad (20)$$

$$B_{ik}^{\text{surv}} = W_{ii} z_i^{-k} \bar{\beta}_k \quad (21)$$

#### 4.4 Combined Coefficients and EBNM Problem

The total quadratic form in  $l_{ik}$  is  $-\frac{1}{2}A_{ik}l_{ik}^2 + B_{ik}l_{ik}$ , where:

$$A_{ik} = \sum_{j=1}^p \tau_j \overline{f_{jk}^2} + W_{ii} \overline{\beta_k^2} \quad (22)$$

$$B_{ik} = \sum_{j=1}^p \tau_j R_{ij}^{-k} \bar{f}_{jk} + W_{ii} z_i^{-k} \bar{\beta}_k \quad (23)$$

**Dimensions:**  $A_{ik}$  is an  $n$ -vector (varies across samples  $i$  via  $W_{ii}$ ). The first term  $\sum_j \tau_j \overline{f_{jk}^2}$  is a scalar (shared across all  $i$ ); the second term  $W_{ii} \overline{\beta_k^2}$  is sample-specific.

##### EBNM Problem

For each sample  $i = 1, \dots, n$ , solve:

$$x_i = \frac{B_{ik}}{A_{ik}}, \quad s_i = \frac{1}{\sqrt{A_{ik}}} \implies (\hat{g}_{l_k}, \hat{q}_{l_k}) = \text{EBNM}(\mathbf{x}, \mathbf{s})$$

The EBNM solver returns:  $\bar{l}_{ik} = \mathbb{E}_{\hat{q}}[l_{ik}]$  and  $\bar{l}_{ik}^2 = \text{Var}_{\hat{q}}(l_{ik}) + \bar{l}_{ik}^2$ .

##### Code Mapping (V2.R)

```
# Lines 310-321: L update for factor k
A_L <- sum(Tau * EF2[, k]) + w * EBeta2[k] # n-vector
A_L <- pmax(A_L, 1e-10) # [A3] floor

B_L_gen <- as.vector(R_k %*% (Tau * EF[, k])) # n-vector
B_L_surv <- w * z_no_k * EBeta[k] # n-vector
B_L <- B_L_gen + B_L_surv

res_L <- ebnm(x = B_L/A_L, s = 1/sqrt(A_L),
             prior_family = "point_normal")
EL[, k] <- res_L$posterior$mean # l_bar
EL2[, k] <- res_L$posterior$sd^2 + res_L$posterior$mean^2 # E[l^2]
```

**Efficiency note:** `R_k %*% (Tau * EF[,k])` computes  $\sum_j \tau_j R_{ij}^{-k} \bar{f}_{jk}$  as a single matrix-vector product ( $n \times p$  times  $p \times 1$ ), avoiding the creation of an intermediate  $n \times p$  matrix from `sweep()`.

## 5 Update for $q_{f_k}$ — Biological Factors

$F$  appears **only** in the genomics term of the likelihood (not in the survival model), so the  $F$ -update is identical to the standard EBMF update.



## 5.1 Derivation

Using the same partial residual  $R_{ij}^{-k}$  from Eq. (13) and taking the expectation over  $q_l$  (using updated values from the  $L$ -step):

$$A_{jk} = \sum_{i=1}^n \tau_j \bar{l}_{ik}^2 = \tau_j \sum_{i=1}^n \bar{l}_{ik}^2 \quad (24)$$

$$B_{jk} = \sum_{i=1}^n \tau_j R_{ij}^{-k} \bar{l}_{ik} = \tau_j \sum_{i=1}^n R_{ij}^{-k} \bar{l}_{ik} \quad (25)$$

**Dimensions:**  $A_{jk}$  is a  $p$ -vector. Since we assume column-specific precision ( $\tau_j$  shared across all  $i$ ), the scalar  $\sum_i \bar{l}_{ik}^2$  factors out and  $A_{jk} = \tau_j \cdot (\text{scalar})$ .

### EBNM Problem

For each feature  $j = 1, \dots, p$ , solve:

$$x_j = \frac{B_{jk}}{A_{jk}} = \frac{\sum_i R_{ij}^{-k} \bar{l}_{ik}}{\sum_i \bar{l}_{ik}^2}, \quad s_j = \frac{1}{\sqrt{A_{jk}}} \implies (\hat{g}_{f_k}, \hat{q}_{f_k}) = \text{EBNM}(\mathbf{x}, \mathbf{s})$$

Note that  $x_j$  does not depend on  $\tau_j$  (it cancels in  $B/A$ ), but  $s_j = 1/\sqrt{\tau_j \sum_i \bar{l}_{ik}^2}$  does: higher-precision features get tighter pseudo-standard-deviations, contributing more signal.

### Code Mapping (V2.R)

```
# Lines 334-340: F update for factor k
sum_EL2_k <- sum(EL2[, k]) # scalar
A_F <- pmax(Tau * sum_EL2_k, 1e-10) # p-vector [A3]
B_F <- Tau * as.vector(t(R_k) %*% EL[, k]) # p-vector

res_F <- ebnm(x = B_F/A_F, s = 1/sqrt(A_F),
              prior_family = "point_normal")
EF[, k] <- res_F$posterior$mean
EF2[, k] <- res_F$posterior$sd^2 + res_F$posterior$mean^2
```

**Note:** `t(R_k) %*% EL[,k]` computes  $\sum_i R_{ij}^{-k} \bar{l}_{ik}$  as a  $p \times n$  times  $n \times 1$  product.

## 6 Update for $q_{\beta_k}$ — Survival Coefficients

$\beta$  appears only in the survival term, so the relevant ELBO component is:

$$\mathcal{F}(q_{\beta}, g_{\beta}) = \mathbb{E}_{q_l, q_{\beta}} [\log P(t, \delta \mid L, \beta)] + \sum_k \mathbb{E}_{q_{\beta_k}} \left[ \log \frac{g_{\beta_k}}{q_{\beta_k}} \right] \quad (26)$$

## 6.1 Expanding Over the Posterior of $L$

Starting from the Taylor surrogate (Eq. 11):

$$\mathbb{E}_{q_l, q_\beta} \left[ \sum_i -\frac{1}{2} W_{ii} (z_i - \eta_i)^2 \right], \quad \eta_i = \sum_k l_{ik} \beta_k \quad (27)$$

The key expansion is:

$$\mathbb{E}_{q_l} \left[ \left( \sum_k l_{ik} \beta_k \right)^2 \right] = \left( \sum_k \bar{l}_{ik} \beta_k \right)^2 + \sum_k \text{Var}_q(l_{ik}) \beta_k^2 \quad (28)$$

This follows from mean-field independence ( $l_{ik} \perp l_{ik'}$  under  $q$ ) and the identity  $\mathbb{E}[l^2] = \text{Var} + (\mathbb{E}[l])^2$ . Substituting into the objective and rearranging:

$$\begin{aligned} \mathcal{F}(q_\beta, g_\beta) &= \mathbb{E}_{q_\beta} \left[ \sum_i -\frac{1}{2} W_{ii} \left( z_i - \sum_k \bar{l}_{ik} \beta_k \right)^2 \right] \\ &\quad - \mathbb{E}_{q_\beta} \left[ \sum_k \frac{1}{2} \left( \sum_i W_{ii} \text{Var}_q(l_{ik}) \right) \beta_k^2 \right] + \text{KL terms} \end{aligned} \quad (29)$$

## 6.2 Coordinate-Ascent for $\beta_k$

Fixing all  $\beta_{k'}, k' \neq k$  and collecting terms in  $\beta_k$ :

- From term 1:  $-\frac{1}{2} [\sum_i W_{ii} \bar{l}_{ik}^2] \beta_k^2 + [\sum_i W_{ii} z_i^{-k} \bar{l}_{ik}] \beta_k$
- From term 2:  $-\frac{1}{2} [\sum_i W_{ii} \text{Var}_q(l_{ik})] \beta_k^2$

Combining and using  $\overline{l_{ik}^2} = \text{Var}_q(l_{ik}) + \bar{l}_{ik}^2$ :

$$\boxed{A_k = \sum_{i=1}^n W_{ii} \overline{l_{ik}^2}} \quad (30)$$

$$\boxed{B_k = \sum_{i=1}^n W_{ii} z_i^{-k} \bar{l}_{ik}} \quad (31)$$

**Dimensions:**  $A_k$  and  $B_k$  are both *scalars* (summed over all  $n$  samples for a single factor  $k$ ).

**Error-in-variables correction.** Using  $\overline{l_{ik}^2}$  (which includes  $\text{Var}_q(l_{ik})$ ) in  $A_k$  rather than  $\bar{l}_{ik}^2$  naturally inflates the effective noise variance, preventing  $\beta_k$  from overfitting to uncertain loadings. This is the Bayesian analogue of errors-in-variables regression.

**Reuse of  $z_i^{-k}$ .** The partial working response  $z_i^{-k} = z_i - \sum_{k' \neq k} \bar{l}_{ik'} \bar{\beta}_{k'}$  is the **same** quantity computed in Section 4.1 for the  $L$ -update. It can be reused because it depends only on  $k' \neq k$  quantities.

### EBNM Problem

For factor  $k$ , solve the EBNM problem with a *single* pseudo-observation:

$$x_k = \frac{B_k}{A_k}, \quad s_k = \frac{1}{\sqrt{A_k}} \quad \implies \quad (\hat{g}_{\beta_k}, \hat{q}_{\beta_k}) = \text{EBNM}(x_k, s_k)$$

The EBNM solver estimates the prior  $g_{\beta_k}$  from this single observation. While a single data point limits the information available for prior estimation, the point-normal family has only two parameters (mixing proportion  $\pi$  and variance  $\sigma^2$ ), and the shrinkage is mainly driven by  $s_k$ . See Section 10 for further discussion.

### Code Mapping (V2.R)

```
# Lines 356-361: Beta update for factor k
A_Beta <- max(sum(w * EL2[, k]), 1e-10)      # scalar [A3]
B_Beta <- sum(w * z_no_k * EL[, k])          # scalar

res_Beta <- ebnm(x = B_Beta/A_Beta, s = 1/sqrt(A_Beta),
                 prior_family = "point_normal")
EBeta[k] <- res_Beta$posterior$mean
EBeta2[k] <- res_Beta$posterior$sd^2 + res_Beta$posterior$mean^2
```

**z\_no\_k reuse:** The variable `z_no_k` computed at line 295 is reused here at line 357. No recomputation is needed.

## 6.3 Connection to Weighted Least Squares

The first term of Eq. (29) has the form of a weighted regression:

$$z_i = \sum_k \bar{l}_{ik} \beta_k + \epsilon_i, \quad \epsilon_i \sim \mathcal{N}(0, W_{ii}^{-1})$$

The WLS estimator (ignoring priors and variance correction) would be  $\hat{\beta} = (\bar{L}^\top W \bar{L})^{-1} \bar{L}^\top W \mathbf{z}$ . The coordinate-ascent EBNM update generalises this by: (i) accounting for uncertainty in  $L$  via  $\bar{l}_{ik}^2$ , and (ii) applying EBNM shrinkage priors.

## 7 Update for $\tau$ — Noise Precision

### 7.1 Objective

The ELBO components involving  $\tau$  are:

$$\mathcal{F}(\tau) = \frac{1}{2} \sum_{ij} [\log \tau_j - \tau_j \bar{R}_{ij}^2] + C \quad (32)$$

Note the **minus sign** before  $\tau_j \bar{R}_{ij}^2$  (see REVISED.tex correction [R7]).

## 7.2 Expected Squared Residuals

Under the mean-field assumption, the expected squared residual is:

$$\bar{R}_{ij}^2 := \mathbb{E}_{q_i, q_f} \left[ \left( Y_{ij} - \sum_k l_{ik} f_{jk} \right)^2 \right] = \underbrace{\left( Y_{ij} - \sum_k \bar{l}_{ik} \bar{f}_{jk} \right)^2}_{\text{squared residual at posterior means}} + \underbrace{\sum_k \left[ \overline{l_{ik}^2 f_{jk}^2} - \bar{l}_{ik}^2 \bar{f}_{jk}^2 \right]}_{\text{variance inflation}} \quad (33)$$

**Derivation of the variance term:**

$$\mathbb{E} \left[ \left( \sum_k l_{ik} f_{jk} \right)^2 \right] = \sum_k \mathbb{E}[l_{ik}^2] \mathbb{E}[f_{jk}^2] + \sum_{k \neq k'} \mathbb{E}[l_{ik}] \mathbb{E}[f_{jk}] \mathbb{E}[l_{ik'}] \mathbb{E}[f_{jk'}] \quad (34)$$

$$= \sum_k \overline{l_{ik}^2 f_{jk}^2} + \sum_{k \neq k'} \bar{l}_{ik} \bar{f}_{jk} \bar{l}_{ik'} \bar{f}_{jk'} \quad (35)$$

Meanwhile:

$$\left( \sum_k \bar{l}_{ik} \bar{f}_{jk} \right)^2 = \sum_k \bar{l}_{ik}^2 \bar{f}_{jk}^2 + \sum_{k \neq k'} \bar{l}_{ik} \bar{f}_{jk} \bar{l}_{ik'} \bar{f}_{jk'}$$

Subtracting, the cross-terms cancel, leaving  $\sum_k [\overline{l_{ik}^2 f_{jk}^2} - \bar{l}_{ik}^2 \bar{f}_{jk}^2]$ , which equals  $\sum_k \text{Var}_q(l_{ik}) \mathbb{E}_q[f_{jk}^2] + \bar{l}_{ik}^2 \text{Var}_q(f_{jk}) + \text{Var}_q(l_{ik}) \text{Var}_q(f_{jk})$ .

## 7.3 Column-Specific Precision MLE

Setting  $\partial \mathcal{F} / \partial \tau_j = 0$ :

$$\hat{\tau}_j = \frac{n}{\sum_{i=1}^n \bar{R}_{ij}^2} \quad (36)$$

### Code Mapping (V2.R)

```
# Lines 374-376: Tau update
Var_Term <- (EL2 %*% t(EF2)) - (EL^2 %*% t(EF^2)) # n x p
R2_bar   <- (Y - EL %*% t(EF))^2 + Var_Term         # n x p
Tau      <- n / pmax(colSums(R2_bar), n * 1e-8)    # p-vector
```

#### Matrix operations:

- `EL2 %*% t(EF2)` computes  $\sum_k \overline{l_{ik}^2 f_{jk}^2}$  ( $n \times p$ ).
- `EL^2 %*% t(EF^2)` computes  $\sum_k \bar{l}_{ik}^2 \bar{f}_{jk}^2$  ( $n \times p$ ).
- Their difference is the variance inflation  $\sum_k [\dots]$  in Eq. (33).
- `colSums(R2_bar)` computes  $\sum_i \bar{R}_{ij}^2$  (a  $p$ -vector).

## 8 Complete Algorithm (Pseudocode $\leftrightarrow$ R Code)

### Supervised MF — CAVI Algorithm

**Inputs:**  $Y_{n \times p}$ ,  $(t_i, \delta_i)_{i=1}^n$ ,  $K$ , `max_iter`, `tol`

**Initialize:**

1. SVD of  $Y$ :  $\bar{L} = U\sqrt{D}$ ,  $\bar{F} = V\sqrt{D}$ ;  $\bar{l}^2 \leftarrow \bar{l}^2$ ,  $\bar{f}^2 \leftarrow \bar{f}^2$ .
2. Warm-start  $\bar{\beta}$ : fit `coxph()` on initial loadings.
3.  $\tau_j \leftarrow 1/\text{Var}(Y_{\cdot j})$  for each column  $j$ .

**For**  $b = 1, 2, \dots, \text{max\_iter}$ :

1. **Cox Taylor expansion:**  $\hat{\eta} = \bar{L}\bar{\beta}$ ;  $(u_i, W_{ii}) \leftarrow \text{CoxTaylor}(\hat{\eta}, t, \delta)$ ;  $z_i = \hat{\eta}_i + u_i/W_{ii}$ .
2. **For**  $k = 1, \dots, K$ :
  - (a) **Compute partial residuals:**  $R^{-k}$ ,  $z_i^{-k}$ .
  - (b) **Update**  $q_{l_k}$ :  $A_{ik}, B_{ik} \rightarrow \text{EBNM} \rightarrow \bar{l}_{ik}, \bar{l}_{ik}^2$ .
  - (c) **Update**  $q_{f_k}$ :  $A_{jk}, B_{jk} \rightarrow \text{EBNM} \rightarrow \bar{f}_{jk}, \bar{f}_{jk}^2$ .
  - (d) **Update**  $q_{\beta_k}$ :  $A_k, B_k \rightarrow \text{EBNM} \rightarrow \bar{\beta}_k, \bar{\beta}_k^2$ .
3. **Update**  $\hat{\tau}_j$ :  $\bar{R}_{ij}^2 \rightarrow \hat{\tau}_j = n / \sum_i \bar{R}_{ij}^2$ .
4. **Convergence:** If  $\text{mean}|\Delta L| < \text{tol}$  **and**  $\text{mean}|\Delta \beta| < \text{tol}$ , stop.

**Return:**  $(\bar{L}, \bar{F}, \bar{\beta}, \bar{\beta}^2, \hat{\tau}, \text{history})$ .

### 8.1 Pseudocode-to-Code Mapping

| Algorithm Step                    | V2.R Code                                      | Line(s) |
|-----------------------------------|------------------------------------------------|---------|
| SVD initialisation                | <code>svd(Y, nu=K, nv=K)</code>                | 187–195 |
| Cox warm-start for $\beta$        | <code>coxph(...)</code>                        | 198–212 |
| $\tau$ init from column variance  | <code>1.0 / col_var</code>                     | 217–218 |
| $\hat{\eta} = \bar{L}\bar{\beta}$ | <code>EL %% EBeta</code>                       | 254     |
| $(u, w) = \text{CoxTaylor}$       | <code>calc_cox_taylor(eta, ...)</code>         | 255     |
| $z_i = \hat{\eta}_i + u_i/W_{ii}$ | <code>eta + taylor\$u / taylor\$w</code>       | 256     |
| $R_{ij}^{-k}$                     | <code>Y - Y_hat + outer(EL[,k], EF[,k])</code> | 290–291 |
| $z_i^{-k}$                        | <code>z - eta_no_k</code>                      | 294–295 |
| $A_{ik}$ (loadings)               | <code>sum(Tau*EF2[,k]) + w*EBeta2[k]</code>    | 310     |

| Algorithm Step             | V2.R Code                                                        | Line(s) |
|----------------------------|------------------------------------------------------------------|---------|
| $B_{ik}^{\text{gen}}$      | <code>R_k %*% (Tau * EF[,k])</code>                              | 315     |
| $B_{ik}^{\text{surv}}$     | <code>w * z_no_k * EBeta[k]</code>                               | 316     |
| EBNM for $l_k$             | <code>ebnm(x=B_L/A_L, s=...)</code>                              | 319     |
| $A_{jk}$ (factors)         | <code>Tau * sum_EL2_k</code>                                     | 335     |
| $B_{jk}$                   | <code>Tau * t(R_k) %*% EL[,k]</code>                             | 336     |
| EBNM for $f_k$             | <code>ebnm(x=B_F/A_F, s=...)</code>                              | 338     |
| $A_k$ (survival coeff)     | <code>sum(w * EL2[,k])</code>                                    | 356     |
| $B_k$                      | <code>sum(w * z_no_k * EL[,k])</code>                            | 357     |
| EBNM for $\beta_k$         | <code>ebnm(x=B_Beta/A_Beta, ...)</code>                          | 359     |
| Variance term              | <code>EL2%*%t(EF2) -<br/>EL^2%*%t(EF^2)</code>                   | 374     |
| $\bar{R}_{ij}^2$           | <code>(Y-EL%*%t(EF))^2 + Var_Term</code>                         | 375     |
| $\hat{\tau}_j$             | <code>n / colSums(R2_bar)</code>                                 | 376     |
| ELBO proxy                 | <code>sum(n/2*log(Tau) - ...)</code>                             | 379     |
| $\Delta L$ convergence     | <code>mean(abs(EL - EL_old))</code>                              | 400     |
| $\Delta \beta$ convergence | <code>mean(abs(EBeta -<br/>EBeta_old))</code>                    | 401     |
| Dual check                 | <code>delta_L &lt; tol &amp;&amp; delta_Beta<br/>&lt; tol</code> | 410     |

## 9 Simulation Design

The V2.R script includes a synthetic benchmark to validate the algorithm.

### 9.1 Data Generating Process

1. **Loadings:**  $L_{n \times K} \sim \mathcal{N}(0, 1)$  i.i.d., with  $n = 250$ ,  $K = 5$ .
2. **Factors:**  $F_{p \times K}$  is sparse: for each factor  $k$ , 5% of features are active ( $\sim \mathcal{N}(0, 25)$ ); the remaining 95% are exactly zero. This yields  $p = 1000$  features.
3. **Genomics data:**  $Y = LF^\top + E$ ,  $E_{ij} \sim \mathcal{N}(0, 1)$ .
4. **Survival times:** Weibull baseline with scale  $\lambda = 0.01$ , shape  $\alpha = 1.5$ , and Cox link:

$$T_i = \left( \frac{-\log(U_i)}{\lambda \exp(\eta_i^{\text{true}})} \right)^{1/\alpha}, \quad U_i \sim \text{Uniform}(0, 1)$$

where  $\eta_i^{\text{true}} = L_i^\top \beta^{\text{true}}$ .

5. **True survival coefficients:**  $\beta^{\text{true}} = (1.5, -1.2, 0.8, -0.5, 0.0)$ . Factor 5 ( $\beta_5 = 0$ ) is purely structural (no prognostic value).
6. **Censoring:** Independent exponential censoring with rate 1/50:  $C_i \sim \text{Exp}(1/50)$ . Observed time  $t_i = \min(T_i, C_i)$ ; event  $\delta_i = \mathbb{I}(T_i \leq C_i)$ .

#### Code Mapping (V2.R)

```
# Lines 519-544: sim_data_fn()
L      <- matrix(rnorm(n * k), n, k)           # Step 1
F_mat <- matrix(0, p, k)                       # Step 2
for (i in 1:k) {
  active <- sample(1:p, round(p * 0.05))
  F_mat[active, i] <- rnorm(length(active), 0, 5)
}
Y <- L %*% t(F_mat) + matrix(rnorm(n * p), n, p) # Step 3
Beta      <- c(1.5, -1.2, 0.8, -0.5, 0.0)      # Step 5
eta_true <- as.vector(L %*% Beta)
raw_times <- (-log(runif(n)) / (0.01 * exp(eta_true)))^(1/1.5) # Step 4
cens_times <- rexp(n, rate = 1/50)             # Step 6
```

## 9.2 Expected Results

- **RMSE** should converge near 1.0 (the noise standard deviation).
- **ELBO proxy** should be non-decreasing across iterations.
- **Estimated**  $\beta$  should recover the sign pattern  $(+, -, +, -, 0)$  with magnitudes proportional to truth.
- **Supervised C-index** should exceed Top-5 PCA C-index.
- **Factor 5** ( $\beta_5^{\text{true}} = 0$ ) should have  $\hat{\beta}_5 \approx 0$  (structural factor correctly identified as non-prognostic).

## 10 Implementation Notes

### 10.1 Orthogonalisation Approximation [A2]

When `orthogonalize = TRUE` (default **OFF**), every 10 iterations the code applies an SVD rotation to  $\bar{L}$  for factor identifiability:

$$\begin{aligned}\bar{L} &\leftarrow U_L D_L, & \bar{F} &\leftarrow \bar{F} V_L \\ \bar{l}^2 &\leftarrow \bar{l}^2, & \bar{f}^2 &\leftarrow \bar{f}^2\end{aligned}$$

This resets the second moments to squared means, **discarding posterior variance**. The true second moment  $\bar{l}_{ik}^2 = \text{Var}_q + \bar{l}_{ik}^2 \geq \bar{l}_{ik}^2$  is only recovered at the next EBNM call. Consequences:

- Transient underestimate of posterior uncertainty every 10 iterations.
- The  $\beta$  precision  $A_k$  is temporarily biased low (less shrinkage).
- The  $\tau$  update omits variance inflation (biased high).

These effects are mild and self-correcting. The orthogonalisation is off by default because Point-Normal EBNM already promotes factor distinctness through sparsity.

## 10.2 Taylor-Per-Outer-Iteration Approximation [A6]

By default (`refresh_taylor = FALSE`),  $(z, w)$  are computed once per outer iteration using  $\hat{\eta}$  from the *start* of that iteration. As  $\bar{L}$  and  $\bar{\beta}$  change within the inner  $k$ -loop, the true linear predictor shifts but  $z$  and  $w$  remain stale.

This is the standard IRLS-within-VI approximation: the outer loop restores exactness by re-computing  $(z, w)$  at the next iteration. Setting `refresh_taylor = TRUE` recomputes them at every  $k$ , giving a more accurate but slower inner loop.

## 10.3 Single-Observation EBNM for $\beta$

Each  $\beta_k$  update passes a *single* pseudo-observation  $(x_k, s_k)$  to EBNM. With the point-normal prior family (two parameters:  $\pi_0$  and  $\sigma^2$ ), the solver has limited data for prior estimation. In practice:

- If  $|x_k|/s_k \gg 1$  (strong signal), the posterior mean is close to  $x_k$ .
- If  $|x_k|/s_k \ll 1$  (weak signal), EBNM shrinks toward zero ( $\pi_0 \rightarrow 1$ ), correctly identifying non-prognostic factors.
- The prior is re-estimated at every CAVI iteration, allowing adaptation as the loadings  $L$  stabilise.

## 10.4 Numerical Floors [A3]

All EBNM precision inputs receive a floor of  $10^{-10}$  via `pmax(..., 1e-10)` to prevent:

- Division by zero in  $s = 1/\sqrt{A}$ .
- Infinite pseudo-observations  $x = B/A$  when  $A \approx 0$ .
- Degenerate EBNM solves with unbounded prior variance.

Similarly, the  $\tau$  denominator is floored at  $n \times 10^{-8}$  to prevent  $\tau_j \rightarrow \infty$  for near-zero-variance features.

## A Errata from Original Derivation Document

The following errors were found during a detailed review of the original derivation document (MF\_Derivations\_UpdateAlgo\_2\_12\_26.pdf) and are corrected in MF\_Derivations\_UpdateAlgo\_REVISED.tex.

| ID | Location   | Error in Original                                                                         | Correction                                                             |
|----|------------|-------------------------------------------------------------------------------------------|------------------------------------------------------------------------|
| R1 | Sec. 1A    | Wrong dimension subscripts on $Y, L, F, E$                                                | $Y_{n \times p} = L_{n \times K} F_{K \times p}^\top + E_{n \times p}$ |
| R2 | Eq. 22     | $z_i^{-k}$ defined with $f_{jk'}$ instead of $\beta_{k'}$                                 | $z_i^{-k} = z_i - \sum_{k' \neq k} \bar{l}_{ik'} \bar{\beta}_{k'}$     |
| R3 | Eqs. 32–35 | $f_{jk}^2, \beta_k^2$ (raw squares) where second moments $\mathbb{E}_q[\cdot]$ are needed | Replaced with $\overline{f_{jk}^2}, \overline{\beta_k^2}$              |
| R4 | Eqs. 47–48 | EBNM inputs subscripted $i$ (sample)                                                      | Corrected to $j$ (feature)                                             |
| R5 | Eq. 62     | $\beta_k$ missing square in expansion                                                     | $\sum_k l_{ik}^2 \beta_k^2$                                            |
| R6 | Sec. 6     | $\beta$ update as generic WLS; no EBNM form                                               | Derived $A_k, B_k$ for EBNM                                            |
| R7 | Eq. 83     | Sign error: $\log \tau + \tau \bar{R}^2$                                                  | Corrected to $\log \tau - \tau \bar{R}^2$                              |
| R8 | Eq. 85     | $\bar{l}_{jk}^2$ (wrong index $j$ )                                                       | Corrected to $\bar{l}_{ik}^2$                                          |



## B V1-to-V2 Algorithmic Changes

| ID | Issue in V1              | V1 Behaviour                                                                | V2 Fix                                                                                             |
|----|--------------------------|-----------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------|
| A1 | Stale $\eta$ in $z^{-k}$ | <code>z.no_k</code> computed from start-of-iter $\eta$ only                 | Recompute <code>eta.no_k</code> from current <code>EL</code> , <code>EBeta</code> inside $k$ -loop |
| A2 | Orthogonalisation        | Always ON; resets <code>EL2/EF2</code>                                      | Behind <code>orthogonalize=FALSE</code> flag (default OFF)                                         |
| A3 | No precision floor       | <code>A.L</code> , <code>A.F</code> , <code>A.Beta</code> could be $\leq 0$ | <code>pmax(..., 1e-10)</code> on all three                                                         |
| A4 | Convergence on $L$ only  | Only <code>delta.L</code> checked                                           | Both <code>delta.L</code> and <code>delta.Beta</code> must be $< \text{tol}$                       |
| A5 | No ELBO tracking         | Only RMSE recorded                                                          | <code>elbo.proxy</code> = genomics log-likelihood tracked                                          |
| A6 | Taylor fixed per iter    | $(z, w)$ fixed for all $k$                                                  | <code>refresh.taylor</code> flag for optional per- $k$ refresh                                     |